Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} : Drain Source Voltage.

V_{GS} : Gate Source Voltage.

V_{TH} : Threshold Voltage.

U_t : Thermal Voltage.

n : Weak Inversion Slope factor.

g_m: Transconductance.

Solutions

1. a) State two advantages and two disadvantages of using analogue circuits for signal processing.

[4]

[bookwork]

- Advantage - Computational capabilities arise from the physics of devices, current, voltage characteristics of NMOS, PMOS, capacitors, diodes, KVL, KCL [1]
- Advantage - One Wire represents many bits of information [1]
- Disadvantage - Sensitive to noise and mismatch [1]
- Disadvantage - Signal is not restored at each stage of computation [1]

b) Figure 1.1 shows a circuit which encodes a continuous-time current signal to a different domain. Identify what domain it is, explain its operation using its corresponding domain plots on how the amplitude is encoded; use equations when appropriate.

[6]

It is an elaborated Axon Hillock Circuit used for conversion to spike domain. [1]

It is used to capture the behavior of neurons.
Integrate a current into a capacitor and then reset when the voltage passes the threshold V_{th}.
High gain A and positive feedback configuration make V_{out} increase rapidly, hit the supply V_{DD}.
This generates a spike. V_{mem} then reaches a voltage dependent on the two capacitors.

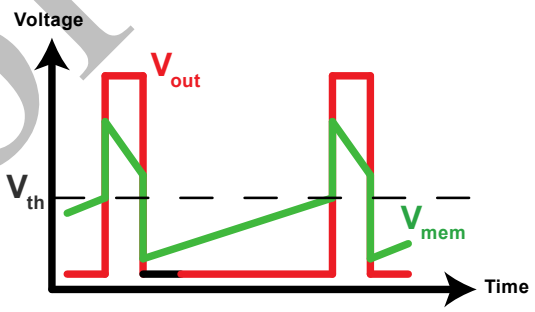
$$\Delta V_{mem} = \frac{C_{fb}}{C_{mem}+C_{fb}} V_{DD}$$

I_r is activated and discharges C_m for t_H until amplifier switches to ground.
Neuron will not fire again for t_L until C_m charges to threshold V_{th}.

$$t_L = \frac{C_{fb}+C_m}{I_{in}} \Delta V_{mem} = \frac{C_{fb}}{I_{in}} V_{DD}$$

$$t_H = \frac{C_{fb}+C_m}{I_r - I_{in}} \Delta V_{mem} = \frac{C_{fb}}{I_r - I_{in}} V_{DD}$$ [3]

Domain plot:



[2]

c) State two advantages of encoding to this domain.

[2]

Any two from the slides. [2]

Solutions

Encoding sensor information in the spike domain has the advantages of :

- **Improved versatility in dynamic range**
  - As the sensory information is encoded in the time domain we are no longer constrained to voltage headroom issues to define the dynamic range. A pulse frequency output can usually vary to 5-6 orders of magnitude which gives plenty of resolution for fine sensor information.
- **Robustness in signal integrity**
  - As the spikes are discrete in amplitude the information transmission has all the benefits which come with encoding information in the digital domain. Furthermore, the signal to noise ratio of any particular pulse frequency output is relatively constant over the entire dynamic range.
- **Inherent compatibility to address-event encoding**
  - This utilises the available bandwidth of communication to the most active events, relaying information only where it occurs and thus conserving power.
- **Technology scalable**
  - Lends itself to modern sub micron processes as it allows operation below 1V, and becomes faster with decreasing transistor sizes.

d) Figure 1.2 shows a simplified three-state implementation of the amplifier used in Figure 1.1. Consider thermal noise and flicker noise only, derive the input referred noise equation at V_{mem} and using it, propose an effective way to reduce the overall input noise. *(You should state the gain expressions using g_m and r_o.)* [8]

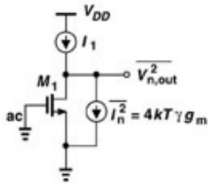
Consider a MOS transistor

Thermal Noise due to drain resistance: $I_n^2 = 4KT(\frac{2}{3}gm)\cdot\Delta f$

Referring Noise sources to the input: $v_n^2 = \frac{I_n^2}{gm^2} = \frac{8}{3}KT\frac{\Delta f}{gm}$

Substituting for current: Strong Inversion - $g_m = 2\sqrt{\beta I_D} = I_D^{0.5}\cdot\sqrt{4\beta}$

Weak Inversion - $g_m = I_D^1 / nU_t$



Input referred thermal noise equation: $v_n^2 = \frac{K_W(p)}{I_D^p}\cdot\Delta f$ **Current dependent !**
Where P=1 for WI and P=0.5 for SI

Flicker Noise: $v_n^2 = \frac{k_f}{C_{ox}WL}\cdot\ln\left(\frac{f_h}{f_l}\right) = \frac{K_f}{A}\cdot\ln\left(\frac{f_h}{f_l}\right)$ **Area dependent ! *A=W.L***

Total MOS Noise: $v_n^2 = \frac{K_w}{I_D^p}\cdot\Delta f + \frac{K_f}{A}\cdot\ln\left(\frac{f_h}{f_l}\right)$ Bandwidth $\Delta f = f_h - f_l$

Gain of each amplifier stage:
$G_n = (g_{mn}\cdot r_{on} // R_{on})$

For M1, refer to the output,
$V_{n1}^2 = \left[\frac{K_w}{I_{D1}^P}\cdot\Delta f + \frac{K_f}{A_1}\cdot\ln\left(\frac{f_h}{f_l}\right)\right]\times(g_{m1}\cdot r_{o1} // R_{o1})(g_{m2}\, r_{o2} // R_{o2})(g_{m3}\, r_{o3} // R_{o3})$

For M2, refer to the output,

$$V_{n2}^2 = \left[\frac{K_w}{I_{D2}^P} \cdot \Delta f + \frac{K_f}{A_2} \cdot \ln\left(\frac{f_h}{f_l}\right)\right] \times \ (g_{m2}\, r_{o2} \,//\, R_{o2})(g_{m3}\, r_{o3} \,//\, R_{o3})$$

For M3, refer to the output,

$$V_{n3}^2 = \left[\frac{K_w}{I_{D3}^P} \cdot \Delta f + \frac{K_f}{A_3} \cdot \ln\left(\frac{f_h}{f_l}\right)\right] \times \ (g_{m3}\, r_{o3} \,//\, R_{o3})$$

Total noise:

$$V_{n-total}^2 = V_{n1}^2 + V_{n2}^2 + V_{n3}^2$$

[8]

Solutions

2. a) State and derive the translinear principle (TLP) for a loop of MOS transistors working in weak inversion.

[bookwork]

In a closed loop containing an even number of forward biased junctions, arranged so that there are an equal number of clockwise facing and counterclockwise facing polarities, the product of the current densities in the clockwise direction is equal to the product of the current densities in the counterclockwise direction.

[2 points for definition]

Number of Clockwise Junctions (Voltage Rises)=Number of Anticlockwise Junctions (Voltage drops)

$$\sum_{n\in CW} V_n = \sum_{n\in CCW} V_n$$

$$V_n = nU_T \ln\left(\frac{I_n}{I_n I_{DO}}\right)$$

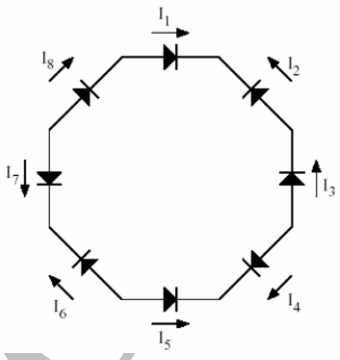
$$\sum_{n\in CW} nU_T \ln\left(\frac{I_n}{I_n I_{DO}}\right) = \sum_{n\in CCW} nU_T \ln\left(\frac{I_n}{I_n I_{DO}}\right)$$

$$\sum_{n\in CW} \ln\left(\frac{I_n}{I_n I_{DO}}\right) = \sum_{n\in CCW} \ln\left(\frac{I_n}{I_n I_{DO}}\right)$$

$$\prod_{n\in CW} \frac{I_n}{I_n I_{DO}} = \prod_{n\in CCW} \frac{I_n}{I_n I_{DO}}$$

$$\prod_{n\in CW} \frac{I_n}{I_n} = I_{DO}^{CW-CCW} \prod_{n\in CCW} \frac{I_n}{I_n}$$

$$\prod_{n\in CW} \frac{I_n}{I_n} = \prod_{n\in CCW} \frac{I_n}{I_n}$$

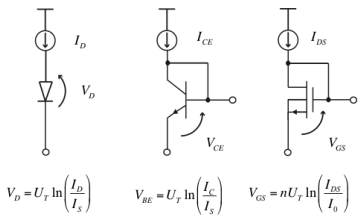


Assumptions: Temperature and saturation currents are all the same [2 points]

[4]

b) Draw three devices which could be used as tranlinear elements justifying your choices.

- The Diode
- The Bipolar Transistor
- The MOS transistor in weak inversion (NB: body effect)



[1 point for each]

[3]

c) Figure 2.1 shows a translinear circuit made up of NMOS transistors biased in weak inversion.

(i) Derive I_{out} as a function of X by applying the translinear principle. Show all necessary steps.

TLP 1: $(1+X)^2 I_0^2 = I_3 \times I_7$

TLP 2: $(1-X)^2 I_0^2 = I_4 \times I_7$

KCL A: $I_4 = 2 \times I_0 - I_3$

Rearranging: $I_7 = (1 + X^2) \times I_0$

KCL B: $I_{OUT} = X^2 \times I_0$

[4]

(ii) What is the function of the circuit shown in Figure 2.1 ?

This is an absolute circuit.

[1]

d) The equation for a vector normaliser circuit is given as:

$$u = \frac{x}{\sqrt{x^2 + y^2}}$$

Synthesise a translinear circuit to provide this function minimsing the circuit where possible. You may only use Type A cells.

Hint: The function may be decomposed into the functions $u = \frac{x}{r}$ and $r = \sqrt{x^2 + y^2}$.

(d) Equation must be synthesised:

Can treat as two separate equations: $r = \sqrt{x^2 + y^2}, u = \frac{x}{r}$

Represent each variable as a ratio of currents:

$$x = \frac{I_x}{I_1}, y = \frac{I_y}{I_1}, r = \frac{I_r}{I_1}, u = \frac{I_u}{I_1}$$

Substitute these into the original equations:

$$\frac{I_u}{I_1} = \frac{I_x/I_1}{I_r/I_1} \quad I_u I_r = I_x I_1$$

$$\frac{I_r}{I_1} = \sqrt{\left(\frac{I_x}{I_1}\right)^2 + \left(\frac{I_y}{I_1}\right)^2} \quad I_r^2 = I_x^2 + I_y^2$$

$$I_r = \underbrace{\frac{I_x^2}{I_r}}_{I_{r1}} + \underbrace{\frac{I_y^2}{I_r}}_{I_{r2}}$$

Derive the translinear circuits which need to be designed:

TLP: $I_{r1} I_r = I_x^2$

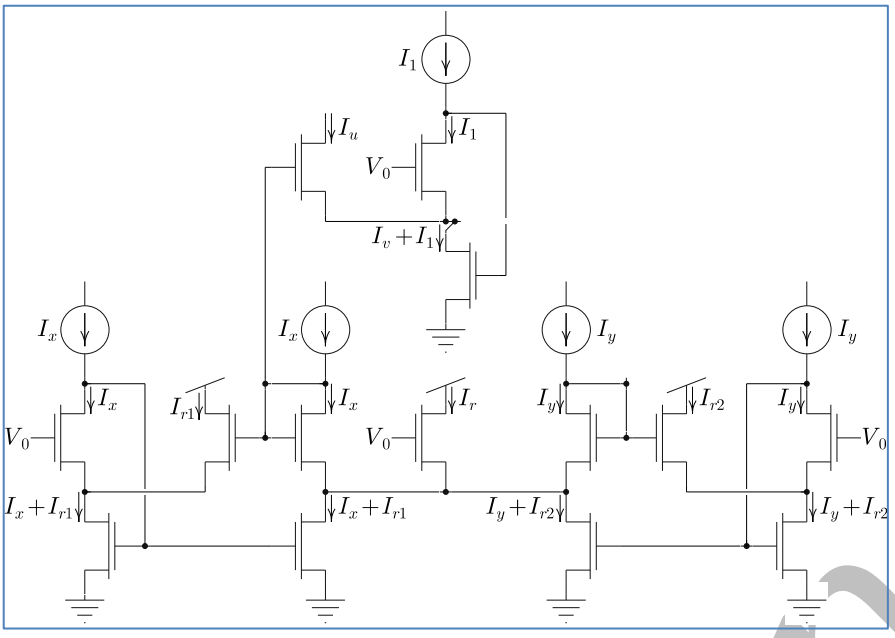
$I_{r2} I_r = I_y^2$

$I_u I_r = I_x I_1$

KCL: $I_r = I_{r1} + I_{r2}$ [4 points for definition of equations]

Final Circuit after removing redundant loops:

Solutions



[4 points for correct circuit]

[8]

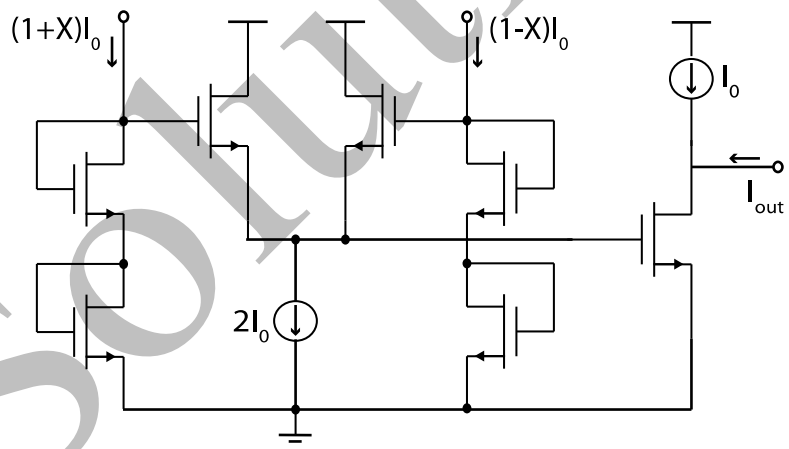


Figure 2.1

Solutions

3. The transfer function of a second order topology has been decomposed into the following state-space equations:

$$\dot{X}_1 = -\frac{w_0}{Q}X_1 + w_0 X_2$$

$$\dot{X}_2 = -w_0 X_1 + w_0 U$$

$$Y_1 = X_1$$

$$Y_2 = X_2$$

whereby Y_1 and Y_2 are the outputs and X_1 and X_2 are the state-variables and U is the inputs.

a. Show that the output Y_1 can be used to implement a second order lowpass transfer function whereas the output Y_2 can be used to implement a "two-pole one-zero" second order transfer function.

New derivation:

$$\left[s + \frac{\omega_0}{Q}\right] X_1(s) - \omega_0 X_2(s) = 0$$

$$\omega_0 X_1(s) + s X_2(s) = \omega_0 U(s)$$

Substituting for $X_2(s)$ (2 points)

$$\frac{Y_1(s)}{U(s)} = \frac{X_1(s)}{U(s)} = \frac{\omega_0^2}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}, \textit{low pass response}$$

Substituting for $X_1(s)$ (2 points)

$$\frac{Y_2(s)}{U(s)} = \frac{X_2(s)}{U(s)} = \frac{\omega_0\left(s + \frac{\omega_0}{Q}\right)}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}, \textit{two pole , one zero response}$$

[Total 4 points]

b. By using the mappings below show how these linear equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U = I_U \exp\left(\frac{V_U}{nU_t}\right)$$

New derivation from Theroy:

Solutions

$$\frac{I_1}{nU_t}\exp\left(\frac{V_1}{nU_t}\right)\dot{V}_1 = -\frac{w_0}{Q}I_1\exp\left(\frac{V_1}{nU_t}\right) + w_0 I_2\exp\left(\frac{V_2}{nU_t}\right)$$

$$\frac{I_2}{nU_t}\exp\left(\frac{V_2}{nU_t}\right)\dot{V}_2 = -w_0 I_1\exp\left(\frac{V_1}{nU_t}\right) + w_0 I_U\exp\left(\frac{V_U}{nU_t}\right)$$

$$C_1\dot{V}_1 = -\frac{C_1 w_0 nU_t}{Q} + C_1 w_0 nU_t\frac{I_2}{I_1}\exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

$$C_2\dot{V}_2 = -C_2 w_0 nU_t\frac{I_1}{I_2}\exp\left(\frac{V_1 - V_2}{nU_t}\right) + C_2 w_0 nU_t\frac{I_U}{I_2}\exp\left(\frac{V_U - V_2}{nU_t}\right)$$

Let C1=C2=C and define $I_w = Cw_0 nU_t$

$$C\dot{V}_1 = -\frac{I_w}{Q} + I_w\frac{I_2}{I_1}\exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

$$C\dot{V}_2 = -I_w\frac{I_1}{I_2}\exp\left(\frac{V_1 - V_2}{nU_t}\right) + I_w\frac{I_U}{I_2}\exp\left(\frac{V_U - V_2}{nU_t}\right)$$

A suitable mapping:

$I_1 = I_2 = I_w$

$I_U = I_0$

Final equations:

$$C\dot{V}_1 = -\frac{I_w}{Q} + I_w\exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

$$C\dot{V}_2 = -I_w\exp\left(\frac{V_1 - V_2}{nU_t}\right) + I_0\exp\left(\frac{V_U - V_2}{nU_t}\right)$$

$$Y_1 = I_w\exp\left(\frac{V_1}{nU_t}\right)$$

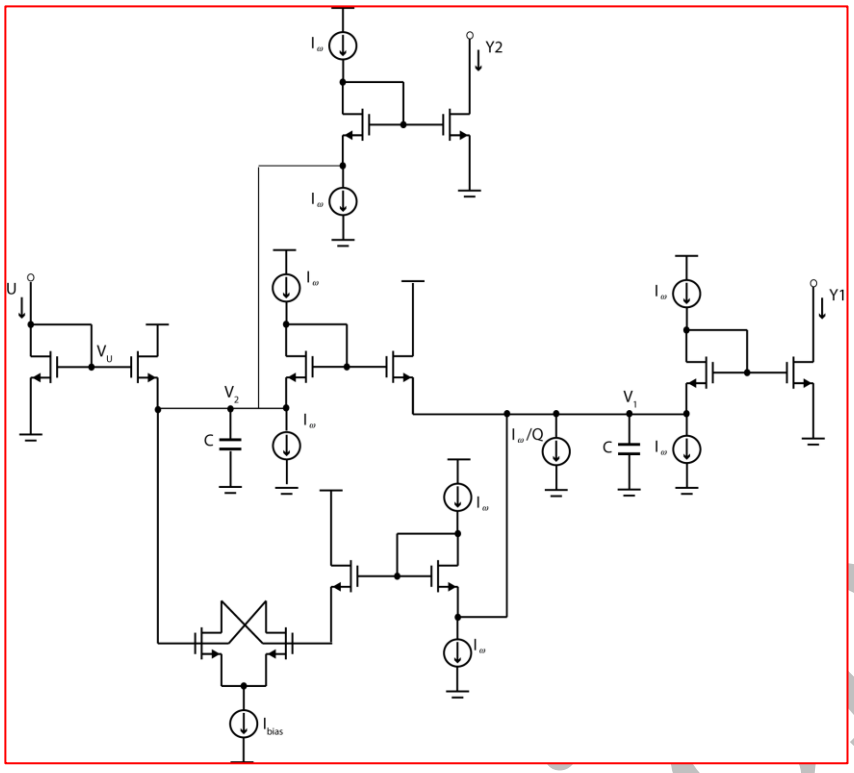
$$Y_2 = I_w\exp\left(\frac{V_2}{nU_t}\right)$$

Total 6 points

c. With these log-domain design equations, sketch a schematic of the final log domain topology which realises the two outputs Y1 and Y2 using weak inversion MOS transistors.

A Solution using the above equations:

Solutions



[Total 8 points]

d. Given the low pass biquad filter has cut-off frequency, f_0 = 4KHz, select a suitable current for I_ω given that the filter capacitor is C = 10 pf , n = 1.23 and U_t = 25mV.

$$I_w = C w_0 n U_t$$

I_ω =10pf x 2π x 4KHz x 1.23 x 25mV =7.7nA

[Total 2 points]

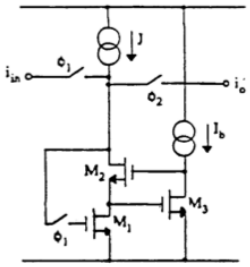
Solutions

4. a) Figure 4.1 depicts a switched current memory cell.

i) State two limitations of the circuit shown in figure 4.1 and draw an improved schematic which compensates for these and has improved output resistance.

The switched current memory cell is susceptible to charge injection from the switch in addition to the drain source voltage variations between the sampling and readout phases which will cause an error in the output current. Increasing the output impedance through cascoding reduces the dependency of I_D on V_{DS}. This is achieved using a regulated cascode: [2]

Regulated Cascode Cell



$$R_o = g_{m2}g_{m3}r_{ds1}r_{ds2}r_{ds3}$$

[2]

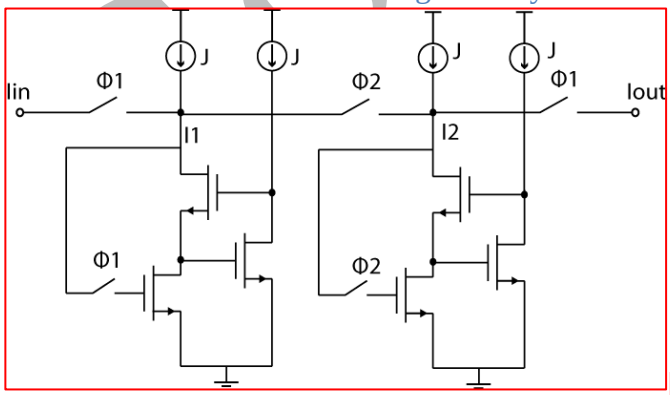
Charge injection effects from switch Φ1 can also be further reduced by operating M1 in the triode region since , $g_{m1}=2\beta V_{DS}$ and therefore error is independent of input signal.

[Total 4 points]

ii) Using your schematic from part 4.a.i, design a memory cell that is non-inverting.

For non-inverting just cascade two cells:

New circuit based on taught theory:



[3]

- On phase Φ_1 of clock period (n-1): $I_1(n-1)=I_{in}(n-1)+J$
- On phase Φ_2 of clock period (n-1): $I_2(n-1)=J-I_{in}(n-1)$
- On phase Φ_1 of clock period (n): $I_{out}(n)=J-I_2(n-1) = I_{in}(n-1)$

[Total 3 points]

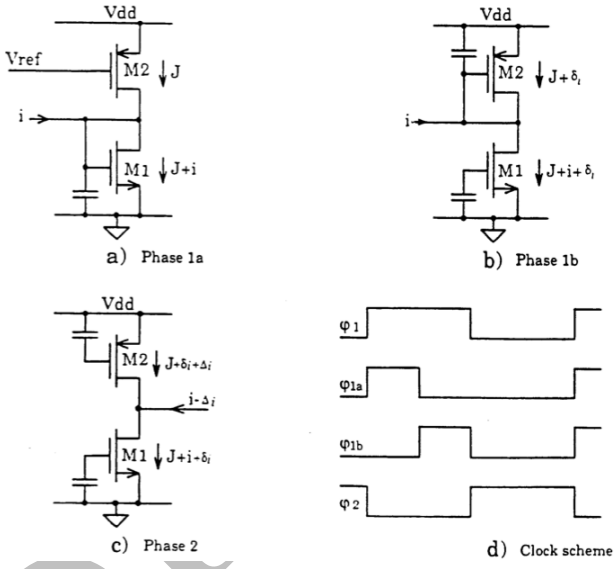
Solutions

iii) Figure 4.2 depicts a switched capacitor memory cell capable of error compensation. Explain its operation, stating the drain currents of M1 and M2 during each phase of the clock and derive the compensated output current.

S^2I memory cell provides error reduction using coarse and fine memory to store the charge injection then subtract it. [1]

S^2I memory cell operation

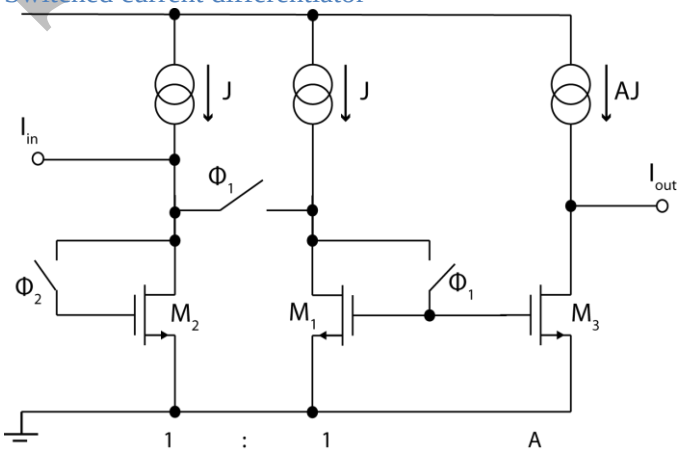
- Phase 1a: $I_{D1}=J+i$, $I_{D2}=J$
- Phase 1b: $I_{D1}=J+i+\delta_1$, $I_{D2}=J+\delta 1$
- Phase 2: $I_{D1}=J+i+\delta_1$, $I_{D2}=J+\delta_1+\Delta_1$
- $I_{out}=I_{D1}-I_{D2}=i-\Delta_1$
- Δ_1 Is an input independent charge injection [1 point for each phase and diagram]



[Total 4 points]

iv) Figure 4.3 depicts a switched current cell. Derive its input/output characteristic and identify which function this block performs. You are required to state the transfer function in the z-domain.

Switched current differentiator



Phase (n-1)
Φ2: $I_{D2}(n-1)=I_{in}(n-1)+J$

Phase (n)
Φ1: $I_{D1}(n)=I_{in}(n) - I_{in}(n-1)+J$
$I_{D3}(n)=AI_{in}(n) - AI_{in}(n-1)+AJ$
$I_{out}(n) = A(I_{in}(n)- I_{in}(n-1)$: This is a difference equation

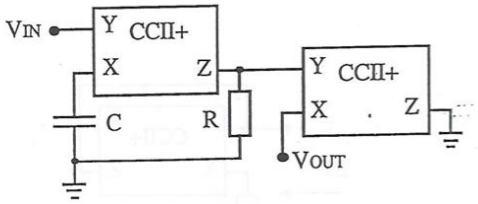
Taking Z-transforms:
$I_{out}(z)= A(I_{in}(z)-I_{in}(z)z^{-1})$

$$\frac{I_{out}(z)}{I_{in}(z)} = A(1 - z^{-1})$$

[Total 5 points]

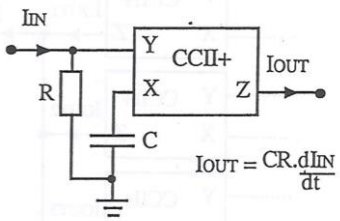
b) This question concerns current conveyors. Design, using CCII+ current conveyors, a voltage mode and a current mode differentiator.

Voltage mode differentiator:



$$V_{out} = RC\frac{dV_{in}}{dt}$$ [2]

Current mode differentiator:



[2]

[Total 4 points]

Solutions

5. This question is on Switched-Capacitor Circuits and techniques.

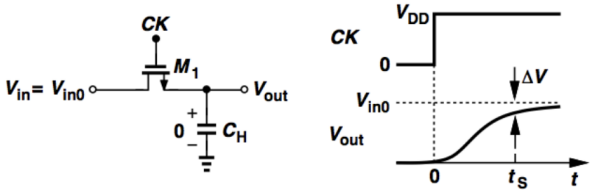
a) (i) Fig.5.1 shows a typical implementation of a switched-capacitor unit cell. Define its voltage range and speed limitations. Use equations as well as $V_{OUT}\ vs.time(t)$ and $R_{ON}\ vs.V_{OUT}$ plots to aid your explanation. *(Note: You may assume an initial condition of the node voltages to aid you explanation.)*

[3]

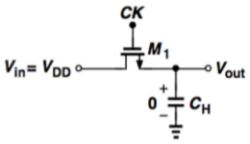
- Channel resistance R_1 is also dependent on the input level V_{in} and yields a greater time constant for higher input levels:

$$R_1 = [\mu_n C_{ox}(W/L)(V_{DD} - V_{in} - V_{TH})]^{-1}$$

$$\tau = C_H \cdot [\mu_n C_{ox}(W/L)(V_{DD} - V_{in} - V_{TH})]^{-1}$$



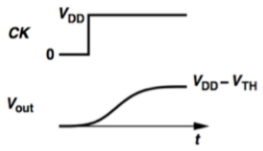
- NMOS Maximum limit: $V_{out} = V_{DD} - V_{TH}$
- When switch is on charging current I_D is:



$$C_H \frac{dV_{out}}{dt} = I_{D1} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{DD} - V_{out} - V_{TH})^2,$$

- Rearranging: $\frac{dV_{out}}{(V_{DD} - V_{out} - V_{TH})^2} = \frac{1}{2}\mu_n \frac{C_{ox}}{C_H}\frac{W}{L}dt,$

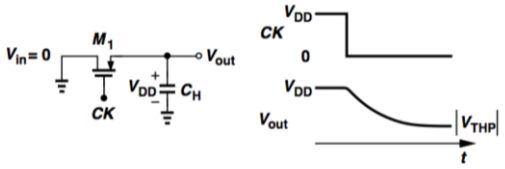
$$\frac{1}{V_{DD} - V_{out} - V_{TH}}\Big|_0^{V_{out}} = \frac{1}{2}\mu_n \frac{C_{ox}}{C_H}\frac{W}{L}t\Big|_0^t,$$



- Integrating this we get: $t \to \infty, V_{out} \to V_{DD} - V_{TH}$.

$$V_{out} = V_{DD} - V_{TH} - \frac{1}{\frac{1}{2}\mu_n \frac{C_{ox}}{C_H}\frac{W}{L}t + \frac{1}{V_{DD} - V_{TH}}}.$$

- PMOS Minimum limit: $|V_{TH}|$
- Same principles can apply to PMOS switches.

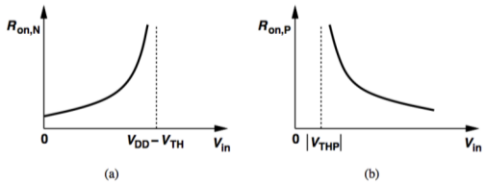


- Need to accommodate greater voltage swings.
- NMOS resistance increases with input voltage.

$$R_{on,N} = [\mu_n C_{ox}(W/L)(V_{DD} - V_{in} - V_{TH})]^{-1}$$

- PMOS resistance decreases with input voltage.

$$R_{on,P} = [\mu_p C_{ox}(W/L)(V_{in} - |V_{THP}|)]^{-1}$$

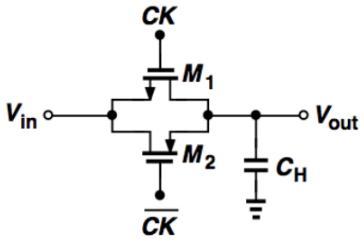


(a) (b)

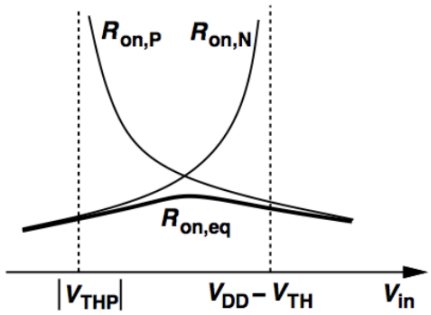
On-resistance of (a) NMOS and (b) PMOS devices as a function of input voltage.

[1] on NMOS range, [1] on PMOS range, [1] on resistance - speed

(ii) Draw an improved design of the switch and state how you may size the transistor(s) to address the previous limitations. [2]



(a)



(b)

$$\begin{aligned} R_{on,eq} &= R_{on,N} || R_{on,P} \\ &= \frac{1}{\mu_n C_{ox}(\frac{W}{L})_N (V_{DD} - V_{in} - V_{THN})} || \frac{1}{\mu_n C_{ox}(\frac{W}{L})_P (V_{in} - |V_{THP}|)} \\ &= \frac{1}{\mu_n C_{ox}(\frac{W}{L})_N (V_{DD} - V_{THN}) + [\mu_n C_{ox}(\frac{W}{L})_N - \mu_p C_{ox}(\frac{W}{L})_P] V_{in} - \mu_p C_{ox}(\frac{W}{L})_P V_{THP}} \end{aligned}$$

- Interestingly we can cancel out input dependent terms if:

$$\mu_n C_{ox}(W/L) = \mu_p C_{ox}(W/L)$$

[2]

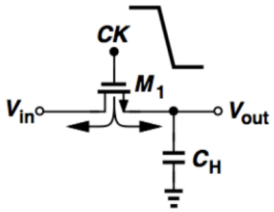
(iii) Explain why charge-injection is a non-ideal effect of using MOSFETs as switches for switched-capacitor unit cells. Draw and state the gain, offset and non-linearity issue due to charge injection when body effect is considered and briefly define three other non-idealities of MOSFET as switches for S-C cells and write down the total voltage variation (ΔV) contributed from all four, including charge-injection. *(Note: You may assume all the channel charges are flowing towards C_H.) (Body effect is given by: $V_{TH} = V_{TH0} + \gamma\sqrt{2\phi_B + V_{SB}} + \gamma\sqrt{\phi_B}$)*

Channel Charge Injection

- When a MOSFET is ON there is a charge in the inversion layer:

$$Q_{ch} = WLC_{ox}(V_{DD} - V_{in} - V_{TH}),$$

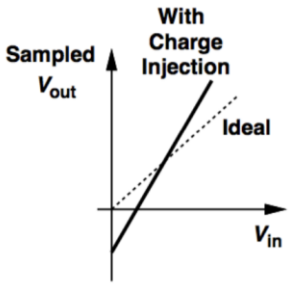
- When the MOSFET turns OFF this charge must exit through the Source and Drain terminals.
- This phenomenon is called 'channel charge injection'.



- Precision is affected.
- Assume all charge is deposited on hold capacitor then:

$$V_{out} \approx V_{in} - \frac{WLC_{ox}(V_{DD} - V_{in} - V_{TH})}{C_H},$$

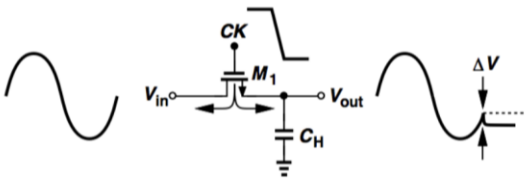
$$V_{out} = V_{in}(1 + \frac{WLC_{ox}}{C_H}) - \frac{WLC_{ox}}{C_H}(V_{DD} - V_{TH}),$$



- Causes a non unity gain equal to $1+WLC_{ox}/C_H$
- Causes a constant offset voltage $-WLC_{ox}(V_{DD}-V_{TH})/C_H$

- Charge injected to the Drain is absorbed by the input source.
- Charge injected to the Source is deposited on C_H.
- This introduces an error on the voltage stored on the capacitor:

$$\Delta V = \frac{WLC_{ox}(V_{DD} - V_{in} - V_{TH})}{2C_H}.$$



- In NMOS switches (in a single-well technology), body effect is an issue since V_B=0 and V_s=V_{in}:

$V_{TH0} + \gamma(\sqrt{2\phi_B + V_{BS}} - \sqrt{2\phi_B})$, and $V_{BS} \approx -V_{in}$, we have

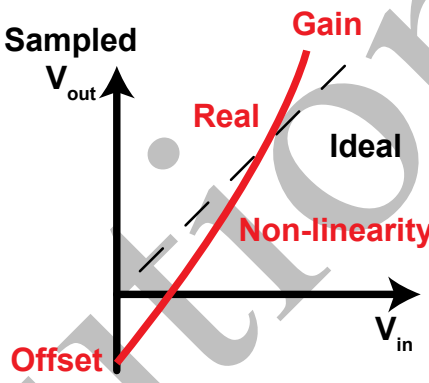
$$\begin{aligned} V_{out} &= V_{in} - \frac{WLC_{ox}}{C_H}(V_{DD} - V_{in} - V_{TH0} - \gamma\sqrt{2\phi_B + V_{in}} - \gamma\sqrt{2\phi_B}), \\ &= V_{in}(1 - \frac{WLC_{ox}}{C_H}) - \gamma\frac{WLC_{ox}}{C_H}\sqrt{2\phi_B + V_{in}} - \frac{WLC_{ox}}{C_H}(V_{DD} - V_{TH0} - \gamma\sqrt{2\phi_B}) \end{aligned} \quad (12.31)$$

- Causes a non-linearity in the input output characteristic.

- Resulting Errors due to Charge injection:
  - **Gain error, DC-offsets** and **Non-linearity**.

[4]

Plot of sampled V_{out} vs V_{in} due to charge injection with body effect considered: [1]



b) (i) Explain the operating principles of the circuits Fig.5.2(a) and (b) and state what improvement the circuit in 5.2(b) achieves compared to that in 5.2(a).

[5]

- Φ1: C_1 charges to V_{in}-V_{os}, C_2 charges to V_{os}, V_{out}=V_{os}
- Φ2: C_1 charges to -V_{os}, C_2 charges to V_{os}-V_{out}
- Charge conservation on node A: Q'_1-Q_1=Q'_2-Q_2
- $C_1(V_{in}-V_{os})-C_1(-V_{os})=C_2V_{os}-C_2(V_{os}-V_{out})$
- $V_{out}=V_{in}C_1/C_2$ independent of V_{os} !
- Output switches from V_{os} to V_{out} requiring the opamp to have high slew rate.

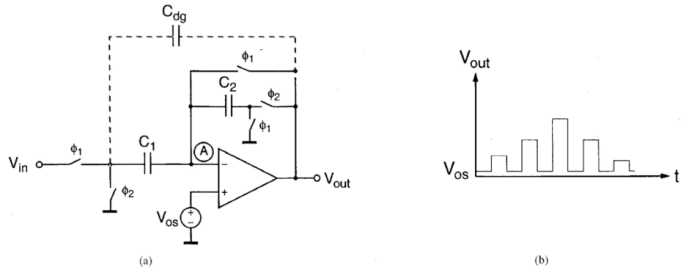


Fig. 28. Offset-compensated SC amplifier: (a) schematic and (b) output signal.

Solutions

- An improved version.
- Does not require resetting of output at each clock period.
- Feedback reset switch is replaced by a S/H branch consisting of C_3 and its two switches.
- Clock phases in parenthesis: inverting amplifier
- Clock phases outside parenthesis: non-inverting amplifier.

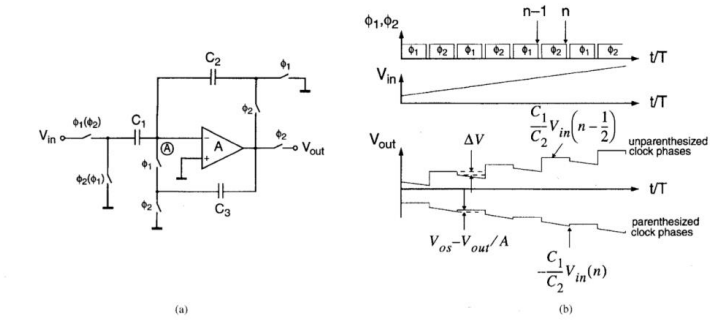


Fig. 30. Offset- and finite-gain compensated SC amplifier: (a) an offset and finite-gain compensated SC amplifier and (b) clock and signal waveforms.

Fig 6.2 shows two offset-free correlated double sampling switched-capacitor non-inverting amplifiers. 6.2b. does not require resetting of output at each clock period to V_{os} thus relaxing the slew rate requirement of the op-amp itself. [2] on identify the functionality and advantage of b, [3] on their principle.

(ii) Derive and explain the functionality of the circuit shown in Fig 5.3.

[3]

Fig.5.3 is an offset-free S-C differentiator. [1]

$\phi_1: Q_1 = (V_{in} - V_{os}) \times C_1;\ Q_2 = (V_o - V_{os}) \times C_2$

$\phi_2: Q_1 = (V'_{in} - V_{os}) \times C_1;\ Q_2 = -V_{os} \times C_2$

Next ϕ_1: due to charge conservation:

$$(V_{in} - V'_{in}) \times C_1 = -V_{o(\phi_1)} \times C_2$$

$$V_{o(\phi_1)} = -\frac{C_1}{C_2} \times (V_{in} - V'_{in}) = -\frac{C_1}{C_2} \times (1 - Z^{-1})$$

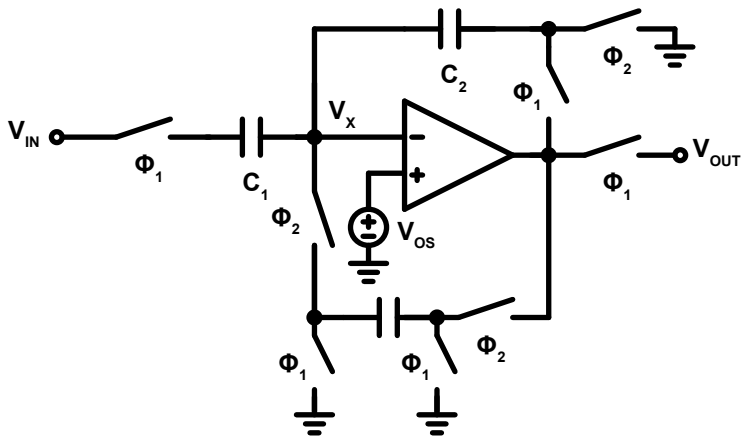
[2]

(iii) Draw a revised circuit of Fig 5.3 which facilitates wideband operation.

[2]

We can use the same method from Fig 5.2(a) to Fig 5.2(b) by replacing the feedback switch to a feedback switched capacitor circuit, so that the output does not reset to V_{os} at ϕ_2. [2]

Solution



Solutions

(i) Draw the block diagram of an auto-zeroing closed-loop amplifier with offset compensation using auxiliary input and explain its operating principle.

[3]

- Store offset at some intermediate node, V_C, instead of amplifier input.
- Choose the voltage gain A_1 of the signal input to be higher than the voltage gain A_2 of the nulling input.
- This will ensure charge injection contribution of S_1 will get divided by A_1/A_2 !
- A_3 is a buffer so as the output is not loaded by C_H.
- $V_{out}=(I_1-I_2)R_L$

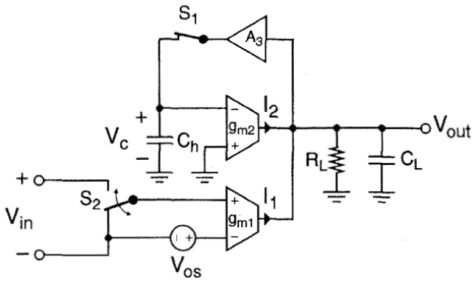


Fig. 23. Offset compensation using an additional offset nulling input.

- Offset will be zero at V_{out} if:
- $I_1=-I_2=g_{m1}xV_{os}=-g_{m2}xV_C$
- Control voltage has to be:
- $V_c=(-g_{m1}/g_{m2})V_{os}$
- Choose g_{m1}/g_{m2} such to ensure V_C is much larger than offset at input to g_{m2}
- Charge injection error appearing at input due to S_1 will get suppressed if $g_{m1}>g_{m2}$

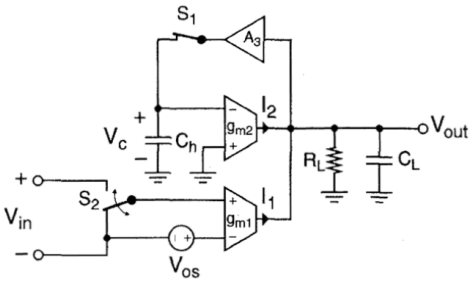


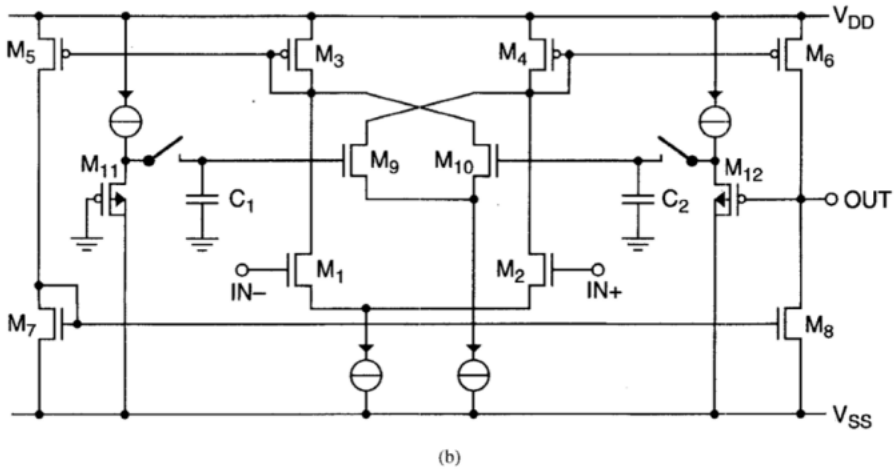
Fig. 23. Offset compensation using an additional offset nulling input.

$$V_{os\text{-}res} \cong -\frac{V_{os}}{A_2A_3} - \frac{A_2}{A_1}\frac{q_{inj}}{C_h}$$
$$= -\frac{V_{os}}{A_2A_3} - \frac{g_{m2}}{g_{m1}}\frac{q_{inj}}{C_h}.$$

[3]

(ii) Complete the circuit in Fig.6.1 to apply the auxiliary-input offset compensation described in 6.a.

[3]



[3]

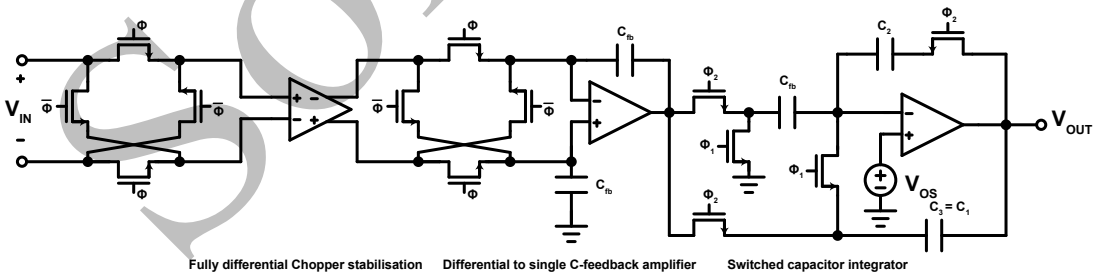
(a) (i) Fig.6.2 shows a typical analogue signal processing chain that continuously amplifies an output signal from a differentially configured sensor pair to a single input ADC. You are responsible to design the missing circuits in the dotted box using the following criteria:

1. At least two-stages of amplification are required followed by an offset-free switched capacitor integrator.

2. The sensor's output has a frequency range of 1–10 Hz, and a signal magnitude of 10 mV. Each op-amp has a 1/f noise amplitude of 10 mV at 10 Hz, and a corner frequency of 1 kHz.

3. You should design the circuit so that the sensor's output is not buried by the input referred noise of your circuit.

4. You are only allowed to use the following symbols as listed in Fig 6.3. You do not need to show the inside schematics of the op-amp itself.



[2] on correct use of chopping switches, [2] on amplifier configurations, [2] on switched capacitor integrator.

[6]

(ii) Explain the method you used to achieve a good noise performance in your sensing system and also explain the operation of the offset-free switched capacitor integrator. You may use frequency spectrum plots to aid your description.

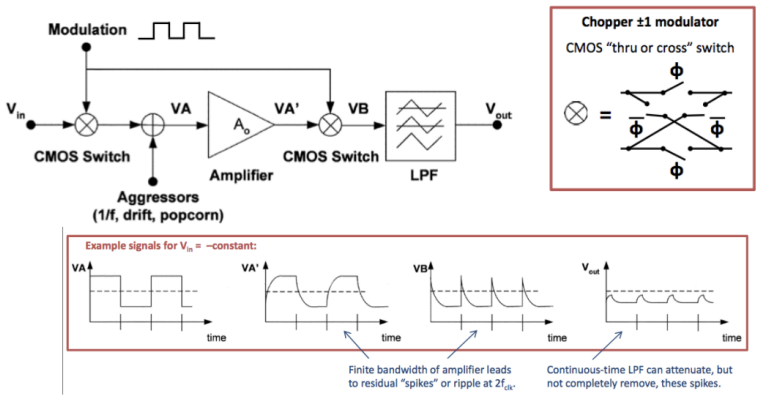
[8]

Bookwork

Solutions

The Chopper Stabilisation Technique

- Upmodulate input signal to frequency above amplifier's 1/*f* noise corner, amplify ac signal, then downmodulate to original frequencies.
- Drift, offset, and 1/*f* noise from amplifier pass only through the second modulator, so they are upmodulated and removed by the low-pass filter.



Frequency Spectrum plots:

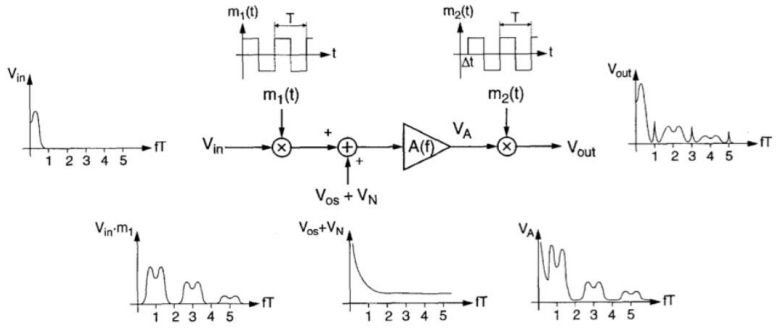


Fig. 9. The chopper amplification principle [19].

[4]

Derivation starting with previous phase (n-1):

- Φ2:
- $Q_1(\Phi2)=C_1(V_{in}(\Phi2)-V_{os})$
- $Q_2(\Phi2)=C_2(V_{os}-V_{out}(\Phi2))$
- $Q_3(\Phi2)=C_3(V_{in}(\Phi2)-V_{out}(\Phi2))$

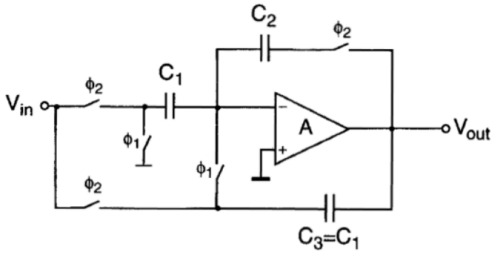


Fig. 41. Offset- and gain-compensated SC integrator.

- Φ1:
- $Q_1(\Phi1)=-C_1V_{os}$
- $Q_2(\Phi1)=Q_2(\Phi2)$ (floating so charge can't change)
- $Q_3(\Phi1)=C_3(V_{os}-V_{out}(\Phi1))$

Φ2: $v_{out}^{\Phi2}(n)=v_{out}^{\Phi2}(n-1)-(C_1/C_2)v_{in}^{\Phi2}(n)$

Φ1: $v_{out}^{\Phi1}(n)=v_{out}^{\Phi2}(n)+v_{os}$

V_{out} at next sample n:

- Φ2:
- $Q_1(\Phi2)-Q_1(\Phi1)=Q_2(\Phi2)-Q_2(\Phi1)$
- $C_1(V_{in}(\Phi2)-V_{os})+C_1V_{os}=C_2(V_{os}-V_{out}(\Phi2))-C_2(V_{os}-V_{out}(\Phi2(n-1))$
- $C_1(V_{in}(\Phi2)=-C_2\,(V_{out}(\Phi2)+V_{out}(\Phi2(n-1)))$
- $V_{out}(n)=V_{out}(n-1)-(C_1/C_2)V_{in}(n)$

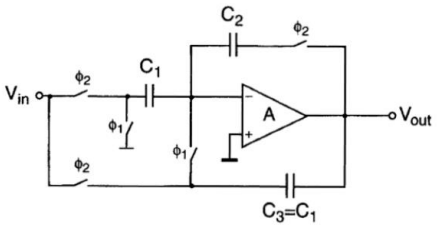


Fig. 41. Offset- and gain-compensated SC integrator.

Φ2: $v_{out}^{\Phi2}(n)=v_{out}^{\Phi2}(n-1)-(C_1/C_2)v_{in}^{\Phi2}(n)$

Φ1: $v_{out}^{\Phi1}(n)=v_{out}^{\Phi2}(n)+v_{os}$

V_{out} at next sample n:

- Φ1:
- $Q_1(\Phi1)-Q_1(\Phi2)=Q_3(\Phi1)-Q_3(\Phi2)$
- $-C_1V_{os}-C_1(V_{in}(\Phi2)-V_{os})=C_3(V_{os}-V_{out}(\Phi1))-C_3(V_{in}(\Phi2)-V_{out}(\Phi2))$
- $-C_1(V_{in}(\Phi2))+C_3(V_{in}(\Phi2))=C_3(V_{os}-V_{out}(\Phi1)-V_{out}(\Phi2))$
- If $C_1=C_2$ then:
- $V_{out}(\Phi1)=V_{out}(\Phi2)+V_{os}$

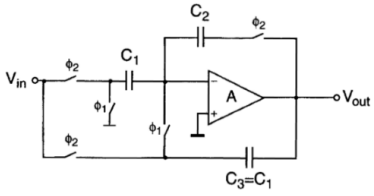


Fig. 41. Offset- and gain-compensated SC integrator.

Φ2: $v_{out}^{\Phi2}(n)=v_{out}^{\Phi2}(n-1)-(C_1/C_2)v_{in}^{\Phi2}(n)$

Φ1: $v_{out}^{\Phi1}(n)=v_{out}^{\Phi2}(n)+v_{os}$

[4]